

BDS FIRST YEAR BIOCHEMISTRY FOUNDATION MODULE:

TOPICS
Introduction to Biochemistry.
Biochemical Composition and Functional Organization of the Cell
Importance of biochemistry in dentistry.
Introduction of Lab Safety Procedures
Cell membrane and its function.
Water structure & Dissociation.
Water, pH and Buffers
Biochemistry Laboratory Instruments and Analytical Techniques
The Extracellular Matrix
Specialized Connective Tissues of Bones, Gums, and Teeth.
Importance of the Extracellular Matrix in Health and Disease.
Preparation of Solutions
Structure & classification of Carbohydrates.
Importance & Functions of Mono, di & oligo-saccharides
Biomedical importance of carbohydrates in health & disease
Biological Samples in Clinical Biochemistry: Collection, Processing, and Applications.
Polysaccharides: Structure, Classification, and Biomedical Importance.
Structure & classification of Amino acids
Importance of amino acids in health & disease
Detection of Carbohydrates
Classification & Functions of Proteins
Structural organization of Proteins
Role of proteins in health & disease
Techniques for identification and separation of Amino Acids
Structure and Functions of Plasma Proteins and Immunoglobulins.
Introduction To Lipids (classification & Functions)
Clinical significance of Plasma Protein and Immunoglobulin Abnormalities
Detection of Protein (Scheme)
Fatty Acids: Structure, classification & Functions.
Structure and Functions of Derived and Compound Lipids
Role of Fatty Acids & Eicosanoids in Health & Disease.
Detection of Protein (Performance)
Nucleotides: Structure, Function, and Biological Roles
Nucleic Acids: Structure and Biological Functions
Cholesterol and Lipid Peroxidation: Mechanisms and Clinical Implications
Proteins Separation techniques (Electrophoresis)
Central Dogma of Molecular Biology.
Minerals in Human Health: Classification and Functions
Vitamins: Classification, Functions, and Health Significance
Test for fat Emulsification